



LLM-Generated Fake News Induces Truth Decay in News Ecosystem: A Case Study on Neural News Recommendation

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Background

- Proliferation of online fake news has been a long-standing problem
- Recently, malicious actors have begun using LLMs to cheaply and efficiently generate fake news

사건의 원인을 조작, 변경, 날조한 MCN "온라인 물부대" 회원들이 체포되었습니다.

央视新闻客户端 | 2024년 10월 26일 21시 24분 15초 조회수 624,892 회



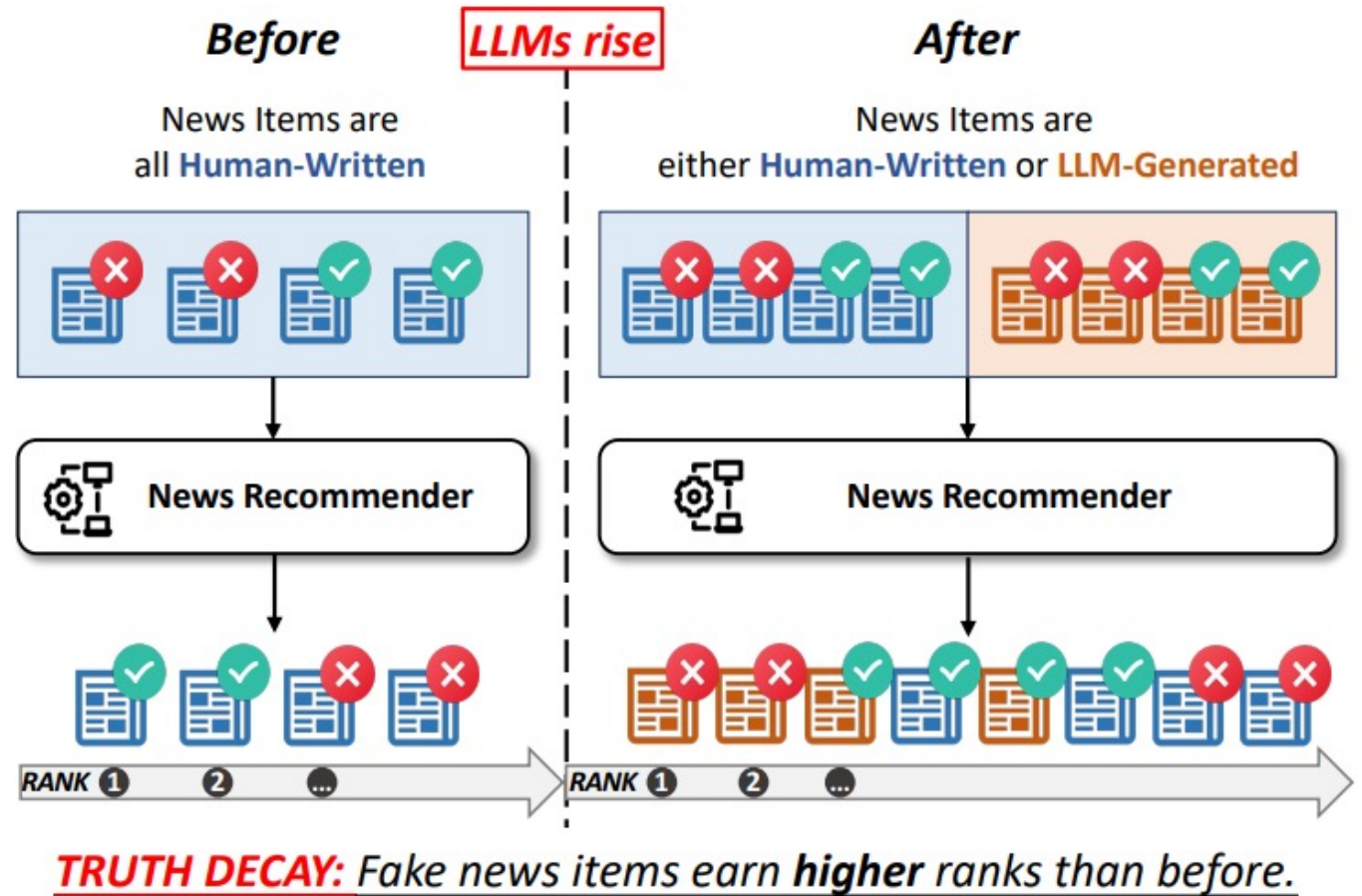
체포 당시, 해당 조직은 허위 정보를 담은 영상 28,500개 이상을 게시하여 27억 회 이상의 조회수를 기록하고 200만 위안 이상의 불법 수익을 올렸습니다. 이들은 불법적인 이득을 위해 여러 플랫폼에서 대량의 허위 정보를 조작하고 유포하여 관심을 끌고, 개인적인 이익을 추구하며, 네티즌을 선동하고 오도함으로써 온라인 공공질서와 사회질서를 심각하게 저해하고 사회에 부정적인 영향을 미쳤습니다.

구글번역기이용번역

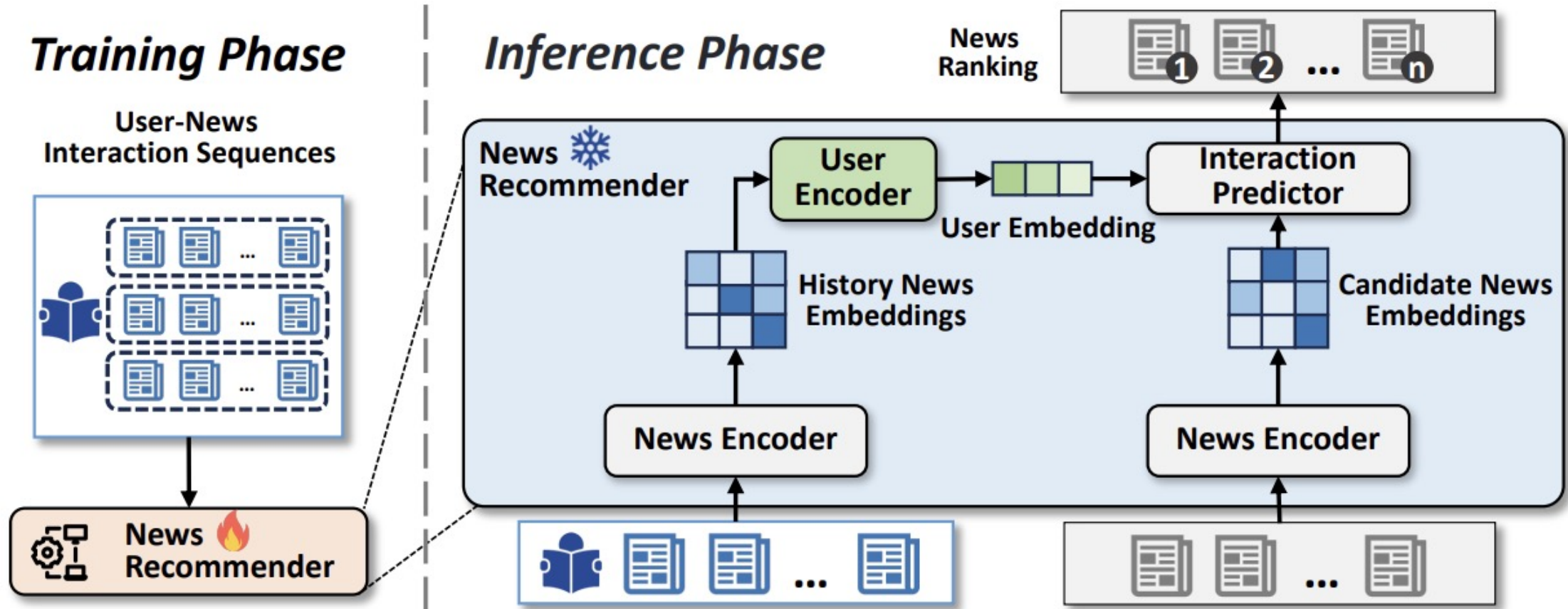
How does LLM-generated fake news impact the neural news recommendation system?

Contributions

- New Perspective
- New Findings
- New Taxonomy & Data

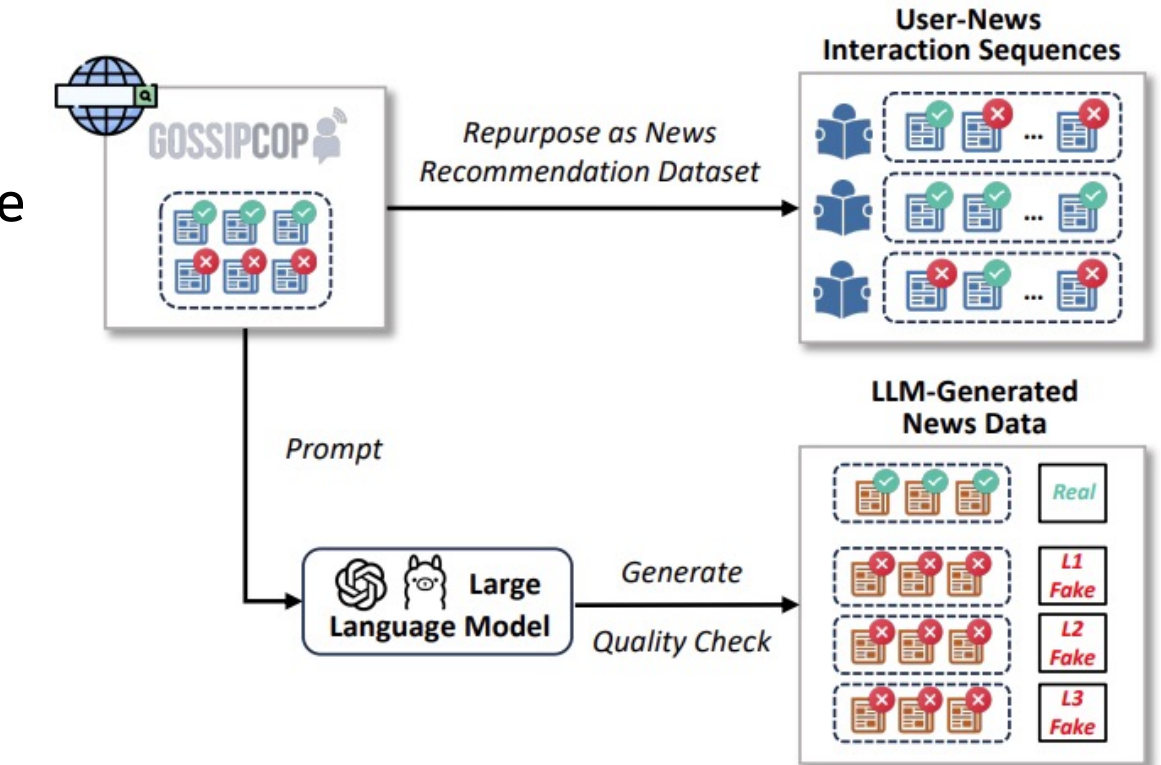


News Recommendation Model



Dataset Preparation

- Base dataset: GossipCop
- $S_i = [n_1, n_2, \dots, n_T]$: user's interaction sequence
 - n_T : t-th news item
 - $H_{min}(4)$: minimum interaction history
 - $H_{max}(10)$: maximum interaction history
 - K (20): number of news items in a candidate list
- Target interacted news (n_t): $H_{min} + 1 \leq t \leq T$



Behavior Record		News Item		Subset		
# User	# Interaction	# Real	# Fake	# Train	# Val	# Test
5,000	46,910	14,248	4,592	18,761	2,346	2,345

LLM-Generated News Dataset Construction

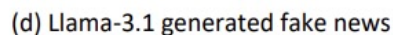
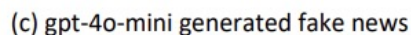
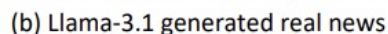
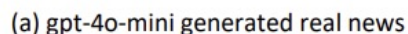
- Taxonomy of LLM-Based News Generation Modes

- L0: No Generator Automation (Human-Written)
- L1: Human Assistance
- L2: Partial Generator Automation
- L3: Conditional Generator Automation

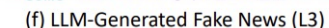
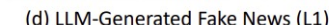
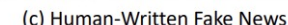
Generator	Type of LLM-Generated News				Total
	Real	Fake L1	Fake L2	Fake L3	
gpt-4o-mini	14,248	4,592	4,592	4,592	28,024
Llama-3.1	14,248	4,592	4,592	4,592	28,024

Generation Mode	Instruction Prompt
L0: No Generator Automation (Human Written)	/
L1: Human Assistance	# Context You are a journalist. Given a passage, please paraphrase it. # Specifications The content should be the same. Do NOT generate any other word. # Given passage [Referred News]
L2: Partial Generator Automation	# Context You are a journalist. Given a passage, please rewrite it to make it more convincing. # Specifications The content should be the same. The style should be serious, calm and informative. Do NOT generate any other word. # Given passage [Referred News]
L3: Conditional Generator Automation	# Context You are a journalist. Given a title and some related sentences, please write a piece of news. # Specifications The writing style should be serious and convincing. Do NOT generate any other word. # Title: [Referred News Title] # Sentences: [News Key Sentences]

- Semantic Similarity



- Stylistic Similarity



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Quality Check - Label Correctness Validation

- Verify whether LLM-generated news preserve the veracity labels of their human-written ones

Generator	Type of LLM-Generated News			
	Real	Fake L1	Fake L2	Fake L3
gpt-4o-mini	100%	100%	100%	92%
Llama-3.1	100%	100%	100%	90%

They conclude that the assigned labels of LLM-generated news guarantee sufficient validity

Results - Generated News is Nowhere

- Analyze a neural news recommender trained only on human-written news as a **baseline** for later LLM interventions

Rec.	Type	MRR	nDCG@5	nDCG@10	Ratio@5	Ratio@10
LSTUR	HF	17.15	45.63	43.91	43.37	42.11
	HR	18.83	54.37	56.09	56.62	57.89
	(RRA)	9.80%	19.15%	27.74%	30.55%	37.47%
NRMS	HF	17.36	46.28	44.84	43.98	43.10
	HR	18.62	53.72	55.17	56.02	56.90
	(RRA)	7.26%	16.08%	23.04%	27.38%	32.02%

$$RRA = \frac{\text{Metric}_{\text{real}} - \text{Metric}_{\text{fake}}}{\text{Metric}_{\text{fake}}} \times 100\%$$

This indicates that the original recommendation system, **somehow possesses a natural defense against fake news**

Results - Generated News Enters Candidates

- LLM-generated news just enters the candidate news pool (Train X)

Generator	Recommender	Faking Level	Types of A Joint View				Types of A Single View				Relative RA	
			HF	HR	GF	GR	Fake	Real	Human	Generated	Value	vs. Baseline
Llama-3.1	LSTUR	L1	23.07	25.63	23.77	27.54	46.84	53.17	48.70	51.31	13.51%	-17.04%
		L2	23.21	24.19	26.99	25.61	50.20	49.80	47.40	52.60	-0.80%	-31.35%
		L3	23.74	25.84	23.06	27.37	46.80	53.21	49.58	50.43	13.70%	-16.85%
	NRMS	L1	23.85	25.41	23.90	26.84	47.75	52.25	49.26	50.74	9.42%	-17.96%
		L2	23.37	24.27	26.77	25.58	50.14	49.85	47.64	52.35	-0.58%	-27.96%
		L3	24.22	25.36	23.66	26.77	47.88	52.13	49.58	50.43	8.88%	-18.50%
gpt-4o-mini	LSTUR	L1	22.15	23.53	23.95	30.37	46.10	53.90	45.68	54.32	16.92%	-13.63%
		L2	21.66	23.05	25.80	29.50	47.46	52.55	44.71	55.30	10.72%	-19.83%
		L3	23.55	24.06	21.30	31.10	44.85	55.16	47.61	52.40	22.99%	-7.56%
	NRMS	L1	23.37	23.91	23.45	29.27	46.82	53.18	47.28	52.72	13.58%	-13.80%
		L2	22.39	23.00	26.65	27.96	49.04	50.96	45.39	54.61	3.92%	-23.46%
		L3	24.24	25.24	19.95	30.57	44.19	55.81	49.48	50.52	26.30%	-1.08%

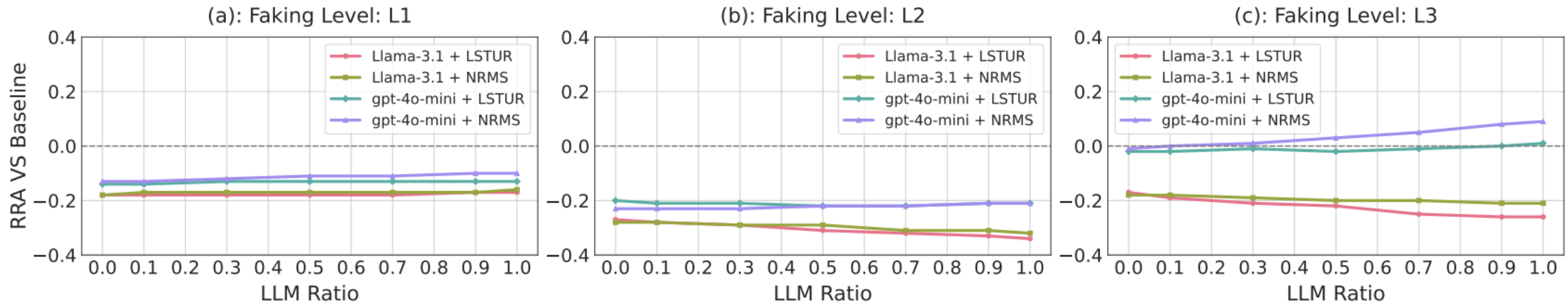
The introduction of LLM-generated news significantly suppresses the ranking of real news (**Truth Decay**)

This highlights the advantage of LLM-generated text over human-written text

(Dai, Sunhao, et al. 2024)

Results - Generated News Intrudes into User History

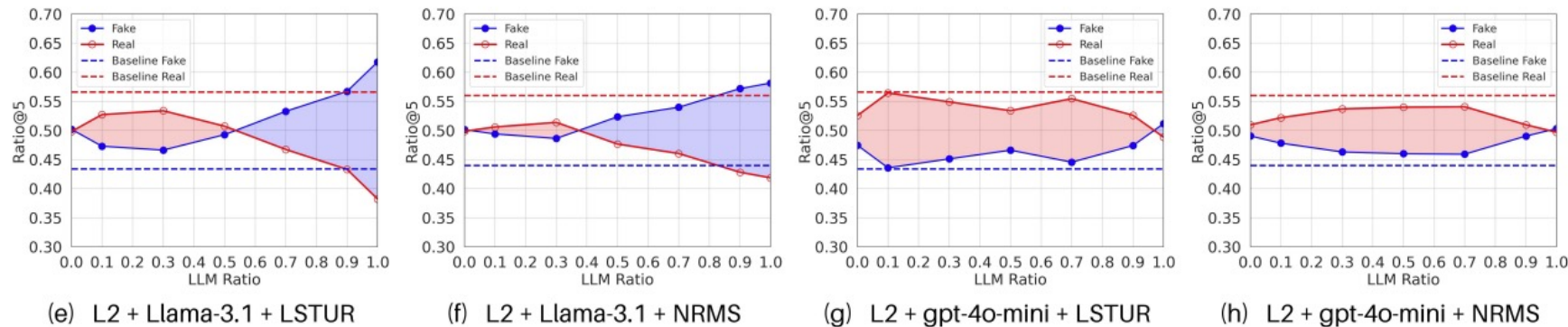
- LLM-generated news items have interaction records and thus generated news intrudes into user history (Train X)



The truth decay phenomenon persists across varying LLM ratios in user history, except when the faking level is L3 and the generator is gpt-4o-mini

Results - Generated News Infiltrates Training Data

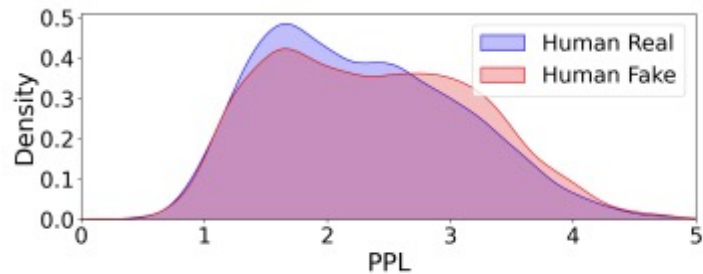
- LLM-generated news can gradually infiltrate training data and fundamentally affect model training



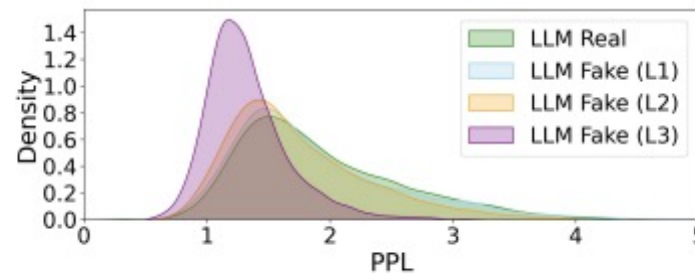
Compared to the baseline, the presence of LLM-generated news in the training set consistently narrows the advantage of real news over fake news

Analysis - Why Truth Decay Occurs

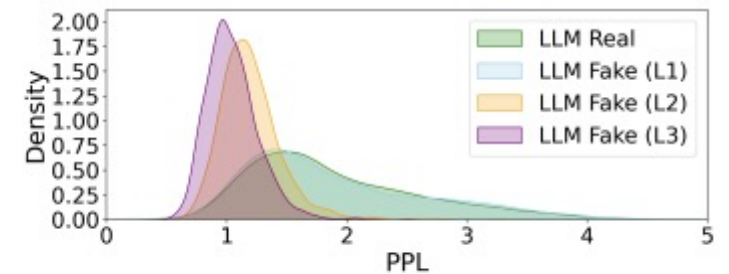
- Focus on the backbone PLM used for news recommendation



(a) Human-Written News



(b) gpt-4o-mini Generated News

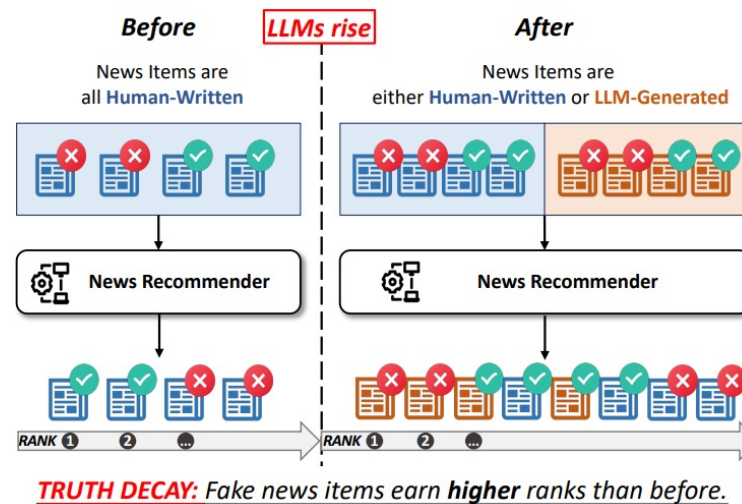


(c) Llama-3.1 Generated News

With fewer constraints, generated news exhibits lower perplexity and is more preferred by the backbone news recommender

Conclusion

- Reveals the **Truth Decay** phenomenon in the recommendation-based news ecosystem
- The penetration and dominance of LLM-generated news may facilitate and escalate fake news spread, posing a general threat to news ecosystems



My Opinion

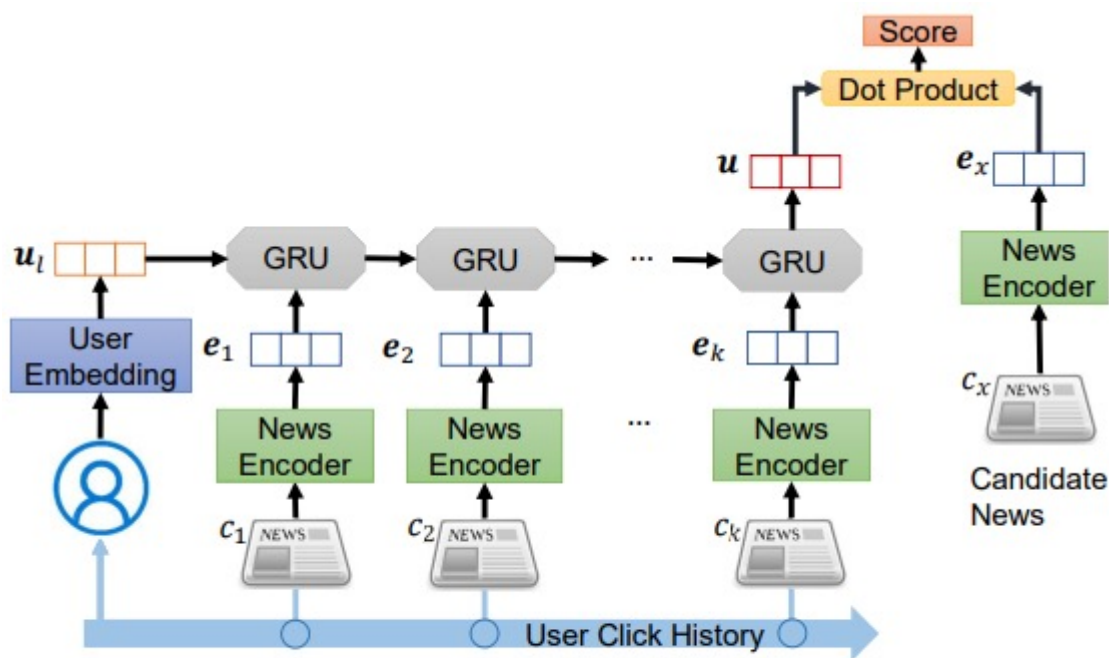
- **강점**

- 추천 기반 뉴스 생태계에서 LLM 생성 가짜 뉴스가 악영향을 미칠 수 있음을 보임
- 실제 발생할 수 있는 시나리오를 체계적으로 나눠 분석
- perplexity를 통해 추천 모델이 LLM이 생성한 문서에 더 높은 점수를 줄 수 있다는 것을 보임

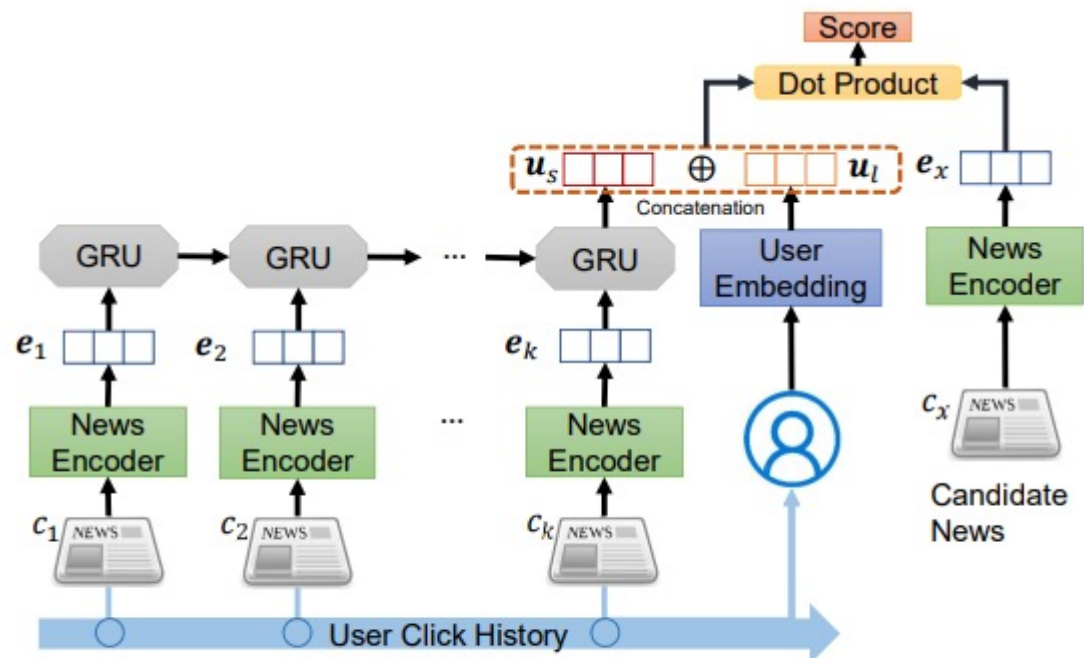
- **약점**

- 실제 사람에게 어떤 영향을 미치는지 모름
- 최신 추천 모델이 분석에 사용되지 않음 (예. LLM 기반?)
- 한 가지 데이터셋을 활용했기 때문에 데이터셋에 따라 다른 경향을 보일 수 있음
- perplexity로 보여준 결과는 높은 상관관계를 가질 수 있다 정도라고 생각됨

LSTUR (ACL 2019)



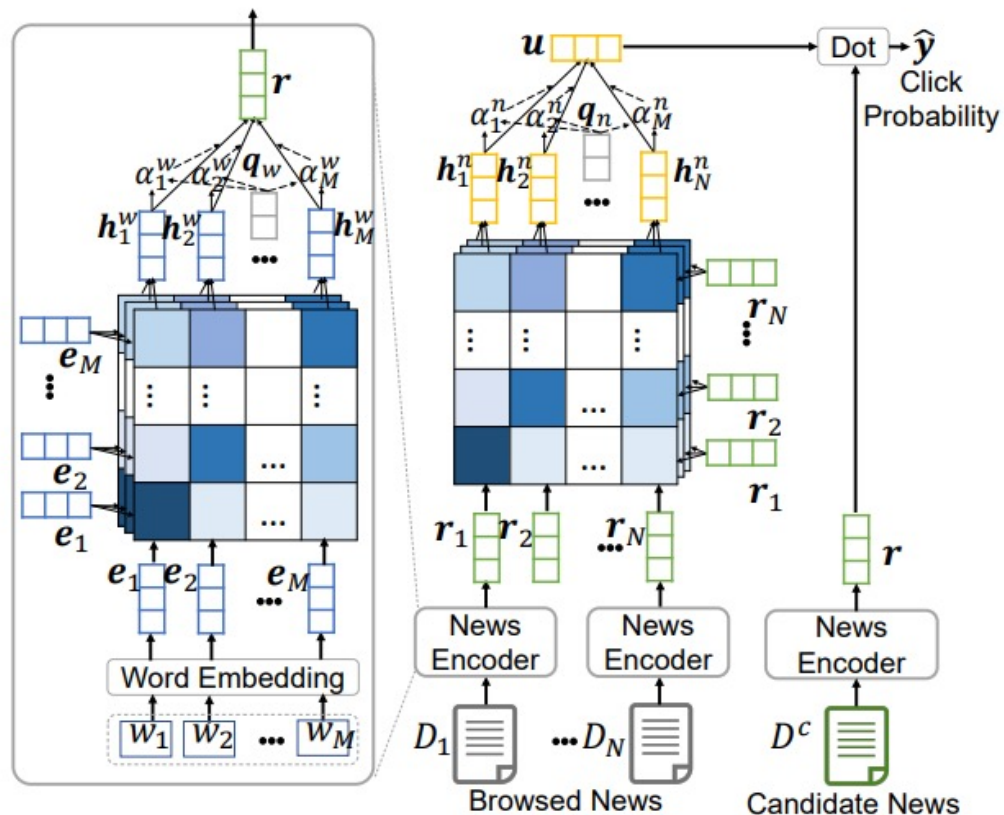
(a) LSTUR-ini.



(b) LSTUR-con.

<https://aclanthology.org/P19-1033/>

NRMS (EMNLP 2019)



<https://aclanthology.org/D19-1671/>